

VFXXX_V5_R4~~R3~~_BATMOBILEFL14

Functional Description

250FL14

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1 Gearbox Status Management - BEV [VFXXX_V5_R4R3]

1.1 Vehicle Function Data

1.1.1 Vehicle Function Area

Powertrain Management

1.1.2 Vehicle Function Group

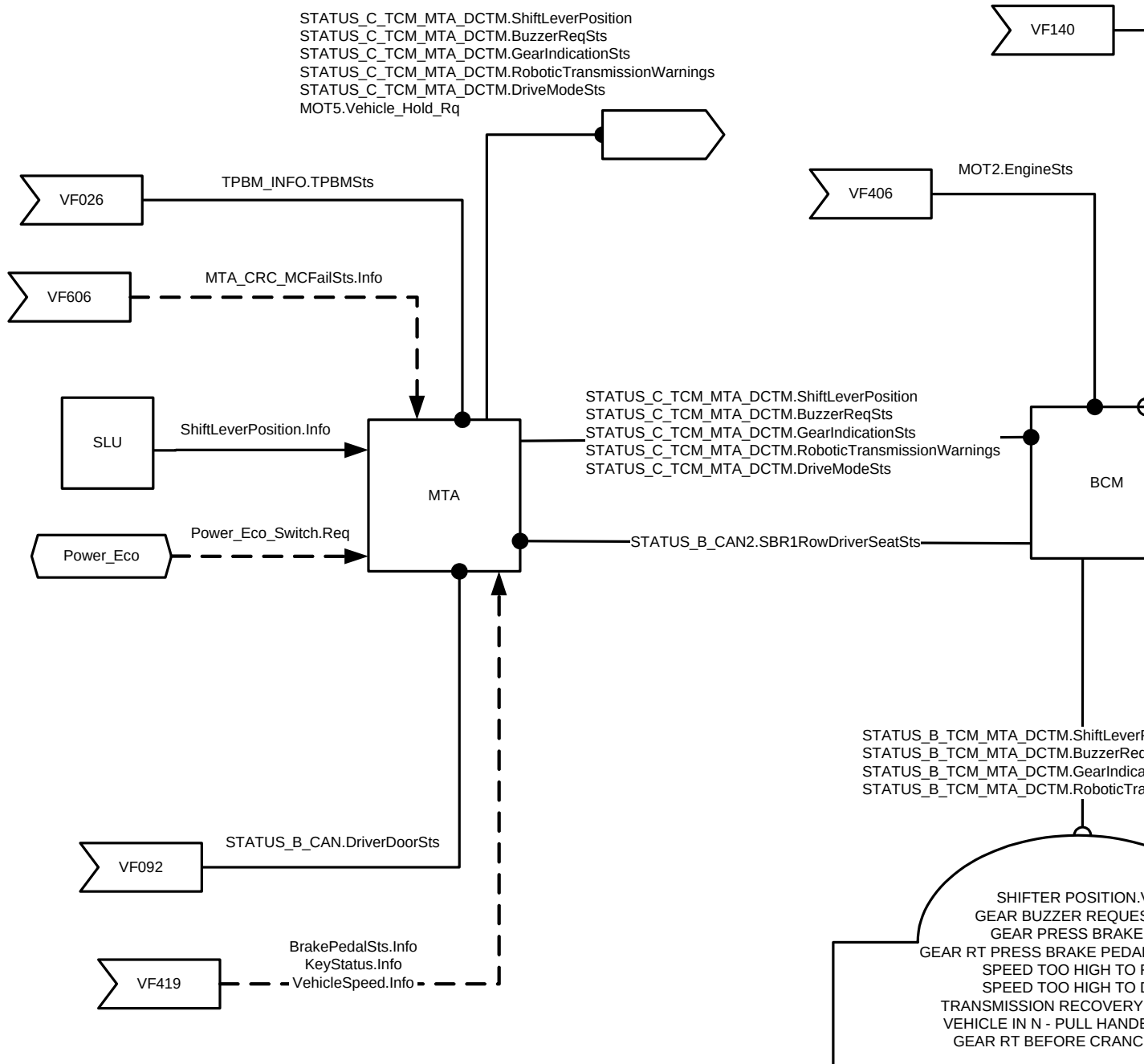
GearBox

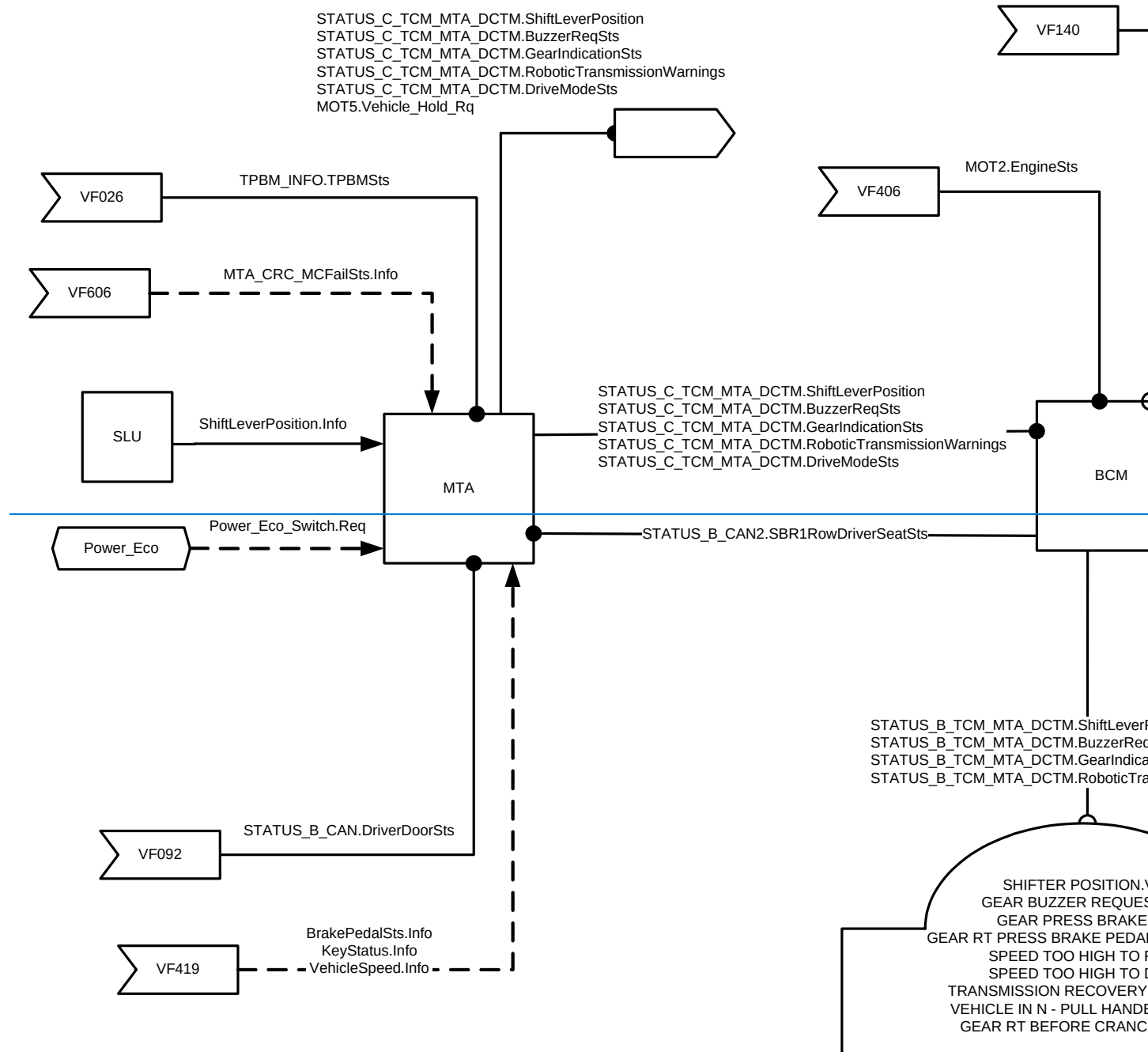
1.2 Revision notes

1.3 Introduction

This function describes transmission indications

1.4 Functional Diagram





1.5 External Interfaces

1.5.1 External Inputs

VF419

VF092

VF140

VF026

VF406

VF606

VF056

1.6 Control Unit

1.6.1 Nodes / ECU

BCM

SLU

IPC

MTA

1.6.2 Nodes / EMCU

1.6.3 Reference

IPC

: acquires commands from CAN;

: indication management for MTA

1.7 I/O

IPC_Display.Gui

Power_Eco

1.8 Information

1.8.1 Hardwire

ShiftLeverPosition.Info

1.8.2 Internal

GearInsertNAndPressBrakeToStart Req

KeyStatus.Info

BrakePedalSts.Info

VehicleSpeed.Info

EcoPower.Req

Power_Eco_Switch.Req

MTA_CRC_MCFailSts.Info

1.8.3 B-CAN

STATUS_B_TCM_MTA_DCTM.GearIndicationSts

STATUS_B_TCM_MTA_DCTM.RoboticTransmissionWarnings

STATUS_B_TCM_MTA_DCTM.BuzzerReqSts

STATUS_B_TCM_MTA_DCTM.ShiftLeverPosition

STATUS_C_CAN.EngineSts

STATUS_B_TCM_MTA_DCTM.DriveModeSts

STATUS_SDM.SBR1RowDriverSeatSts

EDR_INFO.BrakePedalSwitchNOSts

1.8.4 C-CAN

STATUS_C_TCM_MTA_DCTM.GearIndicationSts

This signal is related to the displayed gear indication (if it is blinking or not)

STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings

This signal gives some instruction to the driver in case of malfunctioning (for MTA/DCTM)

STATUS_C_TCM_MTA_DCTM.BuzzerReqSts

It is used to activate the acoustic indicator.

STATUS_C_TCM_MTA_DCTM.ShiftLeverPosition

This signal indicates the actual shift lever position (for TCM or DCTM)

MOT5.Vehicle_Hold_Rq

Vehicle Hold Req

STATUS_B_CAN.DriverDoorSts

This signal indicates the driver door status

STATUS_B_CAN2.SBR1RowDriverSeatSts

It indicates which seats are occupied and if the related seatbelts are fastened.

STATUS_C_TCM_MTA_DCTM.DriveModeSts

This signal indicates the mode selector switch condition selected by driver (Normal, Economy, Sport, Ice,...).

TPBM_INFO.TPBMSts

Transmission Park Brake Status

MOT2.EngineSts

It is the current engine status, according to the following codify:

A. Engine OFF,

B. Engine cranking,

C. Engine ON,

D. Not used In Engine ON status.

1.8.5 LIN

1.8.6 A-BUS

1.8.7 RF

1.8.8 Serial Line

1.9 Indication

GEAR BUZZER REQUEST.V1

Allocated on IPC

GEAR PRESS BRAKE.V6

Allocated on IPC

GEAR RT PRESS BRAKE PEDAL REPEAT.V2

Allocated on IPC

SHIFTER POSITION.V1

Allocated on IPC

SPEED TOO HIGH TO D.V1

Allocated on IPC

SPEED TOO HIGH TO R.V1

Allocated on IPC

TRANSMISSION RECOVERY MODE.V1

Allocated on IPC

MODALITY ECO-POWER.V1

Allocated on IPC

GEAR RT BEFORE CRANCKING.V2

Allocated on IPC

VEHICLE IN N - PULL HANDBRAKE.V1*Allocated on IPC***1.10 Working conditions****1.10.1 Working conditions table**

Working Conditions	Function	Note
Ignition off (+30)	Blinking, buzzer, warning and fail management Gear Changing Request Transmission Status Power_Eco Management Vehicle_Hold_Rq Management Transmission Message management GearInsertNToStart internal managemen Eco Power Interanl management	
Ignition on (+15)	Blinking, buzzer, warning and fail management Gear Changing Request Transmission Status Power_Eco Management Vehicle_Hold_Rq Management Transmission Message management GearInsertNToStart internal managemen Eco Power Interanl management	
Timed with ignition off	Blinking, buzzer, warning and fail management Gear Changing Request Transmission Status Power_Eco Management Vehicle_Hold_Rq Management Transmission Message management GearInsertNToStart internal managemen Eco Power Interanl management	
Cut off during starting	Blinking, buzzer, warning and fail management Gear Changing Request Transmission Status Power_Eco Management Vehicle_Hold_Rq Management Transmission Message management GearInsertNToStart internal managemen Eco Power Interanl management	
Cut off with low battery	Blinking, buzzer, warning and fail management Gear Changing Request Transmission Status Power_Eco Management Vehicle_Hold_Rq Management Transmission Message management GearInsertNToStart internal managemen Eco Power Interanl management	

1.11 Functional requirements

1.11.1 Initialization

1.11.1.1 BCM Management

The BCM shall route the message STATUS_C_TCM_MTA_DCTM as a direct gateway.

The BCM shall route the following signal with this correspondence:

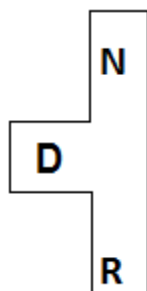
STATUS_B_CAN2.SBR1RowDriverSeatSts = STATUS_SDM.SBR1RowDriverSeatSts

The BCM shall route the following signal with this correspondence:

MOT2.EngineSts = STATUS_C_CAN.EngineSts

1.11.1.2 ESM management

ESM is the HW lever with 3 different stable position R-N-D



<Imagine>

1.11.1.3 MTA management

@KeyStatus.info= keyon :The MTA shall guarantee the CAN update information

@KeyStatus.info= keyoff: The MTA shall guarantee the CAN update information at least for T_SHOW_OFF_MTA

1.11.2 Blinking, buzzer, warning and fail management

1.11.2.1 MTA Algorithm Requirements

1.11.2.1.1 Mismatch table management

@KeySts.info="Keyon_EngineOff": This table represents the cases of Mismatch between the "inserted Gear" and the requested gear :

INPUT	INPUT	INPUT	OUTPUT
ID	ShiftLeverPositon.Info	TPBMSts	Mismatch
1	N	Released	No
2	N	!Released	Yes
3	D	Applied	No
4	D	!Applied	Yes
5	R	Applied	No
6	R	!Applied	Yes

@KeySts.info="Keyon_EngineOn": This table represents the cases of Mismatch between the "inserted Gear" and the requested gear :

INPUT	INPUT	INPUT	INPUT	OUTPUT
ID	ShiftLeverPositon.Info	Direction/Gear of MTA	TPBMSts	Mismatch
1	N	N	Released	No

2	N	N	!Released	Yes
3	N	!N	Released	Yes
4	N	!N	!Released	Yes
5	D	D	Released	No
6	D	D	!Released	Yes
7	D	!D	Released	Yes
8	D	!D	!Released	Yes
9	R	R	Released	No
10	R	R	!Released	Yes
11	R	!R	Released	Yes
12	R	!R	!Released	Yes

1.11.2.1.2 Gear blinking management

MTA manages the blinking on the dashboard with the CAN signal

STATUS_C_TCM_MTA_DCTM.GearIndicationSts, according to these values:

- **STATUS_C_TCM_MTA_DCTM.GearIndicationSts** = Normal to not active the blinking
- **STATUS_C_TCM_MTA_DCTM.GearIndicationSts** = Blinking to active the blinking

@KeyStatus.info= keyon:

MTA shall send **STATUS_C_TCM_MTA_DCTM.GearIndicationSts** = Blinking when:

- it detects mismatch (see the proper table in Mismatch table management depending if the scenario is Keyon_EngineOff or Keyon_EngineOn) after a timeout T_Gear_Mismatch.

MTA shall send **STATUS_C_TCM_MTA_DCTM.GearIndicationSts** =Normal when:

- it doesn't detect mismatch (see the proper table in Mismatch table management depending if the scenario is Keyon_EngineOff or Keyon_EngineOn)

1.11.2.1.3 Buzzer management

MTA could require the buzzer inside with a specific signal **STATUS_C_TCM_MTA_DCTM.BuzzerReqSts** according to these values:

- **STATUS_C_TCM_MTA_DCTM.BuzzerReqSts** = OFF to not active the buzzer
- **STATUS_C_TCM_MTA_DCTM.BuzzerReqSts** = OtherCaseSoundOn to active the buzzer

@KeyStatus.info= keyon:

MTA shall send **STATUS_C_TCM_MTA_DCTM.BuzzerReqSts** = OtherCaseSoundOn when:

- it detects mismatch (see the proper table in Mismatch table management depending if the scenario is Keyon_EngineOff or Keyon_EngineOn) after a timeout T_Gear_Mismatch.

MTA shall send **STATUS_C_TCM_MTA_DCTM.BuzzerReqSts** = OFF when:

- it doesn't detect mismatch (see the proper table in Mismatch table management depending if the scenario is Keyon_EngineOff or Keyon_EngineOn)

1.11.3 Gear Changing Request

1.11.3.1 MTA Algorithm Requirements

1.11.3.1.1 N engagement request

@KeyStatus.info= Keyon_EngineOff:

If the ShiftLeverPosition.Info has a transition in N then

If BrakePedalSts.Info= pressed (it shall be pressed when ShiftLeverPosition has had the transition in N) then

MTA accepts N

Else

MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=

Press_Brake_Pedal_Repeat_Warning for a Tmessage

Endif

Endif

@KeyStatus.info= Keyon_EngineOn:

If the ShiftLeverPosition.Info has a transition in N then

MTA accepts N

Endif

1.11.3.1.2 *R engagement request*

@KeyStatus.info= Keyon_EngineOff:

If the ShiftLeverPosition.Info has a transition in R then

If BrakePedalSts.Info= pressed (it shall be pressed when ShiftLeverPosition has had the transition in R) then
MTA accepts R (but being at engine off it's not ready to drive)

Else

MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
Press_Brake_Pedal_Repeat_Warning for a Tmessage

Endif

Endif

@KeyStatus.info= Keyon_EngineOn:

If ShiftLeverPosition.Info has a transition in R then

If the Previous value of ShiftLeverPosition.Info = D then

If VehicleSpeed <=Rthreshold then

MTA inserts the R (this means that MTA has accepted R)

Else

MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
Vehicle_Speed_Too_High_to_Shift_R for a Tmessage timeout and forces the N

As soon as the Vehiclespeed becomes<=Rthreshold (If ShiftLeverPosition.Info continues to be equal R) the MTA shall:

1)stops to send STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=

Vehicle_Speed_Too_High_to_Shift_R (if the timeout Tmessage is not yet elapsed) and

2)insert R (this means that MTA has accepted R)

Endif

Elseif the Previous value of ShiftLeverPosition = N or No_selection_active_or_available then

If BrakePedalSts.Info= pressed (it shall be pressed when ShiftLeverPosition has had the transition in R) then

If VehicleSpeed.Info <Rthreshold then

MTA inserts the R (this means that MTA has accepted R)

Else

MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
Vehicle_Speed_Too_High_to_Shift_R for a Tmessage timeout and forces the N

As soon as the Vehiclespeed becomes<=Rthreshold (If ShiftLeverPosition.Info continues to be equal R) the MTA shall:

1)stops to send STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=

Vehicle_Speed_Too_High_to_Shift_R (if the timeout Tmessage is not yet elapsed) and

2)insert R (this means that MTA has accepted R)

Else

MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
Press_Brake_Pedal_Repeat_Warning for a Tmessage and forces the N

Endif

Endif

1.11.3.1.3 *D engagement request*

@KeyStatus.info= Keyon_EngineOff:

If the ShiftLeverPosition.Info has a transition in D then

If BrakePedalSts.Info= pressed (it shall be pressed when ShiftLeverPosition has had the transition in D) then
MTA accepts D (but being at engine off it's not ready to drive)

Else

MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
Press_Brake_Pedal_Repeat_Warning for a Tmessage

```

    Endif
  Endif
  @KeyStatus.info= Keyon_EngineOn:
  If ShiftLeverPosition.Info has a transition in D then
    If the Previous value of ShiftLeverPosition = R then
      If VehicleSpeed.Info >=Forwardthreshold then
        MTA shall go in forward direction (this means that MTA has accepted D)
      Else
        MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
        Vehicle_Speed_Too_High_to_Shift_D for a Tmessage timeout and forces the N
        As soon as the Vehiclespeed becomes>=Forwardthreshold (If ShiftLeverPosition.Info continues to be
        equal D) the MTA shall:
        1)stops to send STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
        Vehicle_Speed_Too_High_to_Shift_D (if the timeout is not yet elapsed) and
        2) MTA shall go in forward direction (this means that MTA has accepted D)
      Endif
    ElseIf the Previous value of ShiftLeverPosition = N or or No_selection_active_or_available then
      If BrakePedalSts.Info= pressed (it shall be pressed when ShiftLeverPosition has had the transition in D)
      then
        If VehicleSpeed.Info >=Forwardthreshold then
          MTA shall go in forward direction (this means that MTA has accepted D)
        Else
          MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
          Vehicle_Speed_Too_High_to_Shift_D for a Tmessage timeout and forces the N
          As soon as the Vehiclespeed becomes>=Forwardthreshold (If ShiftLeverPosition.Info continues
          to be equal D) the MTA shall:
          1)stops to send STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
          Vehicle_Speed_Too_High_to_Shift_D (if the timeout is not yet elapsed) and
          2) MTA shall go in forward direction (this means that MTA has accepted D)
        Else
          MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=
          Press_Brake_Pedal_Repeat_Warning for a Tmessage timeout and forces the N
        Endif
      Endif
    Endif
  Endif

```

1.11.4 Transmission Status

1.11.4.1 MTA Algorithm Requirements

1.11.4.1.1 *ShiftLeverPosition signal management*

@KeyStatus.info= Keyon to keyoff transition for all time MTA is on Can Net (for the timeout **T_SHOW_OFF_MTA**)

The MTA shall to manage the STATUS_C_TCM_MTA_DCTM.ShiftLeverPosition according to the following table:

@KeyStatus.info= keyon and KeyStatus.info= keyoff (for the timeout **T_SHOW_OFF_MTA**)

The MTA shall to manage the STATUS_C_TCM_MTA_DCTM.ShiftLeverPosition according to the following table:

INPUT	OUTPUT
ShiftLeverPosition.Info	STATUS_C_TCM_MTA_DCTM.ShiftLeverPosition
R	R
N	N
D	D
!(R, N, D)*	No_selection_active_or_available

*No electrical or plausability fault detected. see Diagnosis paragraph

1.11.5 Power_Eco Management

1.11.5.1 MTA Algorithm Requirements

The MTA acquires the Drive Mode Button. The button shall be managed at KeyOn (both EngineOn and EngineOff)

The Power_Eco button permits to cycle between 3 different mode (Normal, ~~EcoPower~~ and ~~PowerEco~~). The

Sequence of theses mode is Normal - ~~Eco~~ - Power - ~~Eco~~.

At any startup the mode shall be Normal (that is called Predominant Mode) and the MTA sends

STATUS_C_TCM_MTA_DCTM.DriveModeSts=No_program_Selected.

At any keyon to keyoff transition (for the timeout **T_SHOW_OFF_MTA**) the MTA shall send

STATUS_C_TCM_MTA_DCTM.DriveModeSts=No_program_Selected.

At the first button pression (after Tdebounce) the mode switches from Normal to ~~EcoPower~~ and MTA sends

STATUS_C_TCM_MTA_DCTM.DriveModeSts=~~EconomyTracPlus_TowHaul~~.

At the next button pression (after Tdebounce) the mode switches from ~~Eco~~ to Power ~~to Eco~~ and MTA sends

STATUS_C_TCM_MTA_DCTM.DriveModeSts=~~TracPlus_TowHaulEconomy~~.

At the following button pressions (after Tdebounce) the MTA will change the mode following the sequence Normal - ~~Eco~~ - Power - ~~Eco~~. At each mode the MTA shall send the proper value of

STATUS_C_TCM_MTA_DCTM.DriveModeSts

1.11.6 Transmission Parking Brake management

1.11.6.1 MTA Algorithm Requirements

1.11.6.1.1 MOT5.Vehicle_Hold_Rq signal management

1st scenario (Switchoff):

@KeyStatus.info= keyon to keyoff transition:

If VehicleSpeed.Info (last value calculated at Keyon)<TPBM_threshold then

MTA shall send **MOT5.Vehicle_Hold_Rq**= Active (till the MTA is on Can Net)

Else

MTA shall send **MOT5.Vehicle_Hold_Rq**= Not_active and it shall send

STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=Vehicle_In_Neutral (till the MTA is on Can Net)

Endif

2nd scenario (+15 request or cranking request not accomplished)

@KeyStatus.info= keyoff to keyon_engineoff transition

If ShiftLeverPosition.Info*=N and BrakePedalSts.Info*= Active then

MTA shall send **MOT5.Vehicle_Hold_Rq**= Not_active

Elsif ShiftLeverPosition.Info*=N and BrakePedalSts.Info*= Not Active then

If TPBM_INFO.TPBMS* != Released then

MTA shall send **MOT5.Vehicle_Hold_Rq**= active

Elsif TPBM_INFO.TPBMS*=Released then

MTA shall send **MOT5.Vehicle_Hold_Rq**= Not_active

Endif

Elsif ShiftLeverPosition.Info*=D or R

MTA shall send **MOT5.Vehicle_Hold_Rq**= active

Endif

*(MTA shall consider the first acceptable value read after the transition of Keysts)

3rd scenario (cranking request accomplished)

@KeyStatus.info= keyoff or keyon_engineoff to keyon_engineon transition

If ShiftLeverPosition.Info*=N and BrakePedalSts.Info*= Active then

MTA shall send **MOT5.Vehicle_Hold_Rq**= Not_active

Endif

*(MTA shall consider the first acceptable value read after the transition of Keysts)

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4th scenario (lever movement at keyon engine off. This scenario is active only if the 6th scenario is not active):

@KeyStatus.info= keyon_engineoff:

If MTA has accepted N (this means that MTA is in Neutral) then

MTA shall send **MOT5.Vehicle_Hold_Rq**= Not_active (unless what it's described in 6th scenario)

Elseif MTA has accepted D and R then

If VehicleSpeed.Info < TPBM_threshold then

MTA shall send **MOT5.Vehicle_Hold_Rq**= active (Mta Shall send MOT5.Vehicle_Hold_Rq=active till

MTA has accepted other gears)

Else

MTA shall send **MOT5.Vehicle_Hold_Rq**= Not_active and it shall send

STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=Vehicle_In_Neutral

As soon as (if MTA hasn't accepted other gears) the VehicleSpeed.Info < TPBM_threshold the MTA shall:

1) stop to send STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=Vehicle_In_Neutral

AND

2) MTA shall send **MOT5.Vehicle_Hold_Rq**= active. Mta Shall send MOT5.Vehicle_Hold_Rq=active till

MTA has accepted other gears

Endif

Endif

5th scenario (Normal operations, lever movement at keyon engine off. This scenario is active only if the 6th and 7th scenario is not active):

@KeyStatus.info= keyon_engineon:

MTA shall send **MOT5.Vehicle_Hold_Rq**= Not_active (unless what it's described in 6th and 7th scenario)

6th scenario, (driver exit in N. This scenario has priority respect 4th and 5th scenario. ~~(driver exit)~~)

@KeyStatus.info= keyon:

If MTA has accepted N (this means that MTA is in Neutral) then

If STATUS_B_CAN.DriverDoorSts=open OR STATUS_B_CAN2.SBR1RowDriverSeatSts =Seat Belt Not Fasten then

If VehicleSpeed.Info < TPBM_threshold then

MTA shall send **MOT5.Vehicle_Hold_Rq**= active (MTA Shall send MOT5.Vehicle_Hold_Rq=active till the exit conditions are verified*)

Else

MTA shall send MOT5.Vehicle_Hold_Rq= Not_active and it shall send

STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=Vehicle_In_Neutral

As soon as (if the exit conditions are not verified*) the VehicleSpeed.Info < TPBM_threshold the MTA shall:

1) stop to send STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=Vehicle_In_Neutral

AND

2) MTA shall send **MOT5.Vehicle_Hold_Rq**= active. (Mta Shall send

MOT5.Vehicle_Hold_Rq=active till the exit conditions are verified*)

Endif

Endif

Endif

*The exit conditions from 6th scenario are verified if, in the current keycycle: MTA has recognized and accepted a lever movement with STATUS_B_CAN.DriverDoorSts= Closed and STATUS_B_CAN2.SBR1RowDriverSeatSts =Seat Belt fasten

7th scenario (driver exit in D/R. This scenario has priority respect 5th scenario)

@KeyStatus.info= keyon_engineon:

If MTA has accepted D/R (this means that MTA is in Drive or Reverse) then

If STATUS_B_CAN.DriverDoorSts=open AND STATUS_B_CAN2.SBR1RowDriverSeatSts =Seat Belt Not Fasten then

If VehicleSpeed.Info < TPBM_threshold, till the exit conditions* are verified then:

1) MTA shall go to N (this creates mismatch between Lever position and Inserted Gear) AND

2) MTA shall send **MOT5.Vehicle_Hold_Rq**= active

Else

MTA shall continue to manage the current gear

Endif

EndifEndif

*The exit conditions from 7th scenario are verified if, in the current keycycle: MTA has recognized a lever in N (and then there isn't anymore mismatch) with STATUS_B_CAN.DriverDoorSts= Closed OR STATUS_B_CAN2.SBR1RowDriverSeatSts =Seat Belt fasten. (Once this is verified the movement from N to D/R shall be accepted following the strategy described in this document, this means with brake pedal pressed)

1.11.7 Transmission Message management

1.11.7.1 MTA Algorithm Requirements

1.11.7.1.1 *Press_Brake_Pedal_Warning.info signal Management*

@KeyStatus.info= Keyon_engineON if ShiftLeverPosition.Info=N then the MTA shall sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings= Press_Brake_Pedal_warning

1.11.8 GearInsertNAndPressBrakeToStart internal management

1.11.8.1 IPC Algorithm Requirements

If STATUS_B_TCM_MTA_DCTM.ShiftLeverPosition != N OR EDR_INFO.BrakePedalSwitchNOSTs!=Active then

IPC sends GearInsertNAndPressBrakeToStart.Req = Active

Else

IPC sends GearInsertNAndPressBrakeToStart.Req = Not_Active

Endif

1.11.9 EcoPower internal management

1.11.9.1 IPC Algorithm Requirements

IPC shall manage the internal signal EcoPower.Req following this table:

ID	INPUT	OUTPUT
N°	STATUS_B_TCM_MTA_DCTM.DriveModeSts	EcoPower.Req
1	No_programs_available_selected	No_programs_available_selected (default value in case of missing input)
2	Eco	Eco
3	TracPlus_TowHaul	TracPlus

1.12 Indication and text message

1.12.1 Indication and text message

1.12.1.1 IPC - Key Condition Management

"Keyon" (used in the paragraphs below) corresponds to these Key conditions:

- KeyOn_EngineOff [(Hw Key=+15 And STATUS_C_CAN.EngineSts=Engine_OFF) OR (Hw Key=+50 And STATUS_C_CAN.EngineSts=Engine_OFF)]
- Cranking (STATUS_C_CAN.EngineSts=Engine_Cranking)
- KeyOn_EngineOn (Hw Key=+15 And STATUS_C_CAN.EngineSts=Engine_ON)

"Keyoff" (used in the paragraphs below) corresponds to these Key conditions:

- Off (Hw Key=Off)

"**Keyon_EngineOff**" (used in the paragraphs below) corresponds to these Key conditions:

- KeyOn_EngineOff [(Hw Key=+15 And STATUS_C_CAN.EngineSts=Engine_OFF) OR (Hw Key=+50 And STATUS_C_CAN.EngineSts=Engine_OFF)]

@KeyStatus.info= keyon :The IPC shall guarantee the CAN update information

@KeyStatus.info= keyoff:The IPC shall guarantee the CAN update information at least for T_SHOW_OFF_IPC

1.12.1.2 Indication strategy

1.12.1.2.1 Shifter Position indication management

@keyon:

IPC displays SHIFTER POSITION indication according to the STATUS_B_TCM_MTA_DCTM.ShiftLeverPosition AND

If STATUS_B_TCM_MTA_DCTM.GearIndicationSts= Blinking then
IPC displays SHIFTER POSITION blinking

Elsif STATUS_B_TCM_MTA_DCTM.GearIndicationSts= Normal then
IPC displays SHIFTER POSITION in normal mode

Endif

1.12.1.2.2 Gear Buzzer Request indication management

@keyon:

If STATUS_B_TCM_MTA_DCTM.BuzzerReqSts = OtherCaseSoundOn then
IPC activates the GEAR BUZZER REQUEST indication

Endif

1.12.1.2.3 Gear Press Brake indication management

@keyon:

If STATUS_B_TCM_MTA_DCTM.RoboticTransmissionWarnings= Press_Brake_Pedal_Warning then
IPC activates the GEAR PRESS BRAKE indication

Endif

1.12.1.2.4 Gear Press Brake Pedal Repeat indication management

@keyon:

If STATUS_B_TCM_MTA_DCTM.RoboticTransmissionWarnings= Press_Brake_Pedal_Repeat_Warning then
IPC activates the GEAR RT PRESS BRAKE REPEAT indication

Endif

1.12.1.2.5 Speed too high to R indication management

@keyon:

If STATUS_B_TCM_MTA_DCTM.RoboticTransmissionWarnings= Vehicle_Speed_Too_High_to_Shift_R then

IPC activates the SPEED TOO HIGH TO R indication

Endif

1.12.1.2.6 Speed too high to D indication management

@keyon:

If STATUS_B_TCM_MTA_DCTM.RoboticTransmissionWarnings= Vehicle_Speed_Too_High_to_Shift_D then

IPC activates the SPEED TOO HIGH TO D indication

Endif

1.12.1.2.7 Gear Rt Before Crancking indication management

@Keyon_EngineOff:

If GearInsertNAndPressBrakeToStart.Req=Active then

IPC displays GEAR RT BEFORE CRANCKING indication

Endif

1.12.1.2.8 Transmission Recovery mode indication management

@keyon:

If STATUS_B_TCM_MTA_DCTM.RoboticTransmissionWarnings= Trans_Recover_Mode_Warning then

IPC activates the TRANSMISSION RECOVERY MODE indication

Endif

1.12.1.2.9 Modality Eco-Power indication management

@keyon:

IPC shall activates the MODALITY ECO_POWER indication according to EcoPower.Req internal signal

The “Indication” activation and deactivation details are defined in “IPC System Technical Requirement” document.

1.12.1.2.10 Vehicle in N - Pull Handrbake indication management

@keyon and @keyoff (for T_SHOW_OFF_IPC after keyon to keyoff transition)::

If STATUS_B_TCM_MTA_DCTM.RoboticTransmissionWarnings= Vehicle_In_Neutral then

IPC activates the VEHICLE IN N - PULL HANDBRAKE indication

Endif

1.12.1.3 Indication management

Refer to Instrument Panel Cluster documents.

1.13 Diagnosis and recovery**1.13.1 Diagnosis table**

Output Input (Component name)	Diagnosis type (SC-OC etc)	Fault on network (Signal name)	Fault stored (ECU name)	UDM (Detection time) *
ShiftLeverPosition.info	Electrical Diagnosis	n.a.	MTA	See SDF
ShiftLeverPosition.info	Plausability	n.a.	MTA	See SDF
Power_Eco_Switch.Req	Electrical Diagnosis	n.a.	MTA	See SDF
Power_Eco_Switch.Req	Plausability	n.a.	MTA	See SDF
STATUS_B_CAN	Missing Message	n.a.	MTA	See SDF
STATUS_B_CAN2	Missing Message	n.a.	MTA	See SDF
TPBM_INFO	Missing Message	n.a.	MTA	See SDF
MTA_CRC_MCFailSts.Info	Plausability	n.a.	MTA	See SDF
STATUS_SDM	Missing Message	n.a.	BCM	See SDF
STATUS_C_CAN	Missing Message	n.a.	IPC	See SDF
STATUS_B_TCM_MTA_DCTM	Missing Message	n.a.	IPC	See SDF
EDR_INFO	Missing Message	n.a.	IPC	See SDF

* The detection time is defined as the time that is between the fault relevation and its storing.

1.13.2 Diagnosis and recovery description

1.13.2.1 MTA Diagnosis and recovery description

1.13.2.1.1 *ShiftLeverPosition.Info electrical diagnosis description*

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA recognize some Electrical Fault in ShiftleverPosition, then MTA shall:

- set DTC as "validated"
- MTA shall send Motor Fault to IPC (see VF998)
- MTA shall perform this recovery:

If the fault occurs with forward gear inserted then

(MTA maintains the D till KeyStatus becomes KeyOff) AND

MTA sends STATUS_C_TCM_MTA_DCTM.RoboticTransmissionWarnings=Trans_Recover_Mode_Warning
AND

MTA sends STATUS_C_TCM_MTA_DCTM.ShiftLeverPosition=D

Elsif the fault occurs with R/N inserted then

MTA forces the N AND

MTA sends STATUS_C_TCM_MTA_DCTM.ShiftLeverPosition=N

Elsif the fault occurred in previous key cycle then

MTA forces the N AND

MTA sends STATUS_C_TCM_MTA_DCTM.ShiftLeverPosition=N

Endif

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA doesn't recognize any Electrical Fault in ShiftleverPosition, then MTA shall

- set DTC as "devalidated"
- MTA shall stop to send Fail to IPC (VF998)
- MTA shall consider valid the info in ShiftleverPosition.info

1.13.2.1.2 *ShiftLeverPosition.Info Plausability diagnosis description*

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA recognizes some Plausability Fault in ShiftleverPosition, then MTA shall:

- set DTC as "validated"
- MTA shall send Motor Fault to IPC (see VF998)
- MTA shall perform the recovery described in ShiftLeverPosition Electrical fault diagnosis

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA doesn't recognize any Plausability Fault in ShiftleverPosition, then MTA shall

- set DTC as "devalidated"
- MTA shall stop to send Fail to IPC (VF998)
- MTA shall consider valid the info in ShiftleverPosition.info

1.13.2.1.3 *Power_Eco_Switch.Req electrical diagnosis description*

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA recognize some Electrical Fault in Power_Eco_Switch.Reg, then MTA shall:

- set DTC as "validated"
- MTA shall Force STATUS_C_TCM_MTA_DCTM.DriveModeSts=No_program_Selected and goes Normal Mode

In the following "Detection Condition":

- Keyon
- Keyon_engineON

f MTA doesn't recognize any Electrical Fault in Power_Eco_Switch.Reg, then MTA shall

- set DTC as "devalidated"
- MTA shall send STATUS_C_TCM_MTA_DCTM.DriveModeSts coherent with the strategy described in this document

1.13.2.1.4 *Power_Eco_Switch.Reg Plausability diagnosis description*

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA recognizes some Plausability Fault (button pressed for more than TStuck) in Power_Eco_Switch.Reg, then MTA shall:

- set DTC as "validated"
- MTA shall Force STATUS_C_TCM_MTA_DCTM.DriveModeSts=No_program_Selected and goes Normal Mode

In the following "Detection Condition":

- Keyon
- Keyon_engineON

if MTA doesn't recognize any Plausability Fault in Power_Eco_Switch.Reg, then MTA shall

- set DTC as "devalidated"
- MTA shall send STATUS_C_TCM_MTA_DCTM.DriveModeSts coherent with the strategy described in this document

1.13.2.1.5 *STATUS_B_CAN missing message description*

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_B_CAN is not received for a time greater or equal than "MissingMsg_Default_Time" parameter, then mta

- set DTC as "validated"
- MTA shall consider STATUS_B_CAN.DriverDoorSts= Closed

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_C_CAN message is correctly received, then IPC shall:

- set DTC as "devalidated"
- MTA shall consider STATUS_B_CAN.DriverDoorSts received
- MTA shall consider valid the info in ShiftLeverPosition.info

1.13.2.1.6 *STATUS_B_CAN2 missing message description*

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_B_CAN is not received for a time greater or equal than "MissingMsg_Default_Time" parameter, then mta

- set DTC as "validated"

- MTA shall consider STATUS_B_CAN2.SBR1RowDriverSeatSts= Seat Belt Fasten

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_C_CAN message is correctly received, then MTA shall:

- set DTC as "devalidated"
- MTA shall consider STATUS_B_CAN2.SBR1RowDriverSeatSts

1.13.2.1.7 TPBM_INFO missing message description

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If TPBM_INFO is not received for a time greater or equal than "*MissingMsg_Default_Time*" parameter, then mta

- set DTC as "validated"
- MTA shall consider the last value of TPBM_INFO.TPBMSts (in case of no previous value MTA shall consider TPBMSts=Released)

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If TPBM_INFO message is correctly received, then MTA shall:

- set DTC as "devalidated"
- MTA shall consider the last value of TPBM_INFO.TPBMSts

1.13.2.1.8 MTA_CRC_MCFailSts.info Plausability diagnosys description

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA reads MTA_CRC_MCFailSts.info=TPBM_INFO_CRC_Fail or TPBM_INFO_MC_Fail for "*MTA_CRC_MC_wrong_timer*" continuous time then:

- set DTC as "validated"

EVCU implements the same strategy as when it detects TPBM_INFO Missing message

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If MTA doesn't read anymore MTA_CRC_MCFailSts.info=TPBM_INFO_CRC_Fail or TPBM_INFO_MC_Fail then:

- set DTC as "devalidated"
- MTA shall consider the signal in TPBM_INFO

1.13.2.2 BCM Diagnosis and recovery description

1.13.2.2.1 STATUS_SDM missing message description

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_SDM is not received for a time greater or equal than "*MissingMsg_Default_Time*" parameter, then BCM

- set DTC as "validated"
- BCM shall send STATUS_B_CAN2.SBR1RowDriverSeatSts= Seat Belt Fasten

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_SDM message is correctly received, then BCM shall:

- set DTC as "devalidated"
- BCM shall send STATUS_B_CAN2.SBR1RowDriverSeatSts coherent with STATUS_SDM.SBR1RowDriverSeatSts

1.13.2.3 IPC Diagnosis and recovery description

1.13.2.3.1 STATUS_C_CAN missing message description

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_C_CAN is not received for a time greater or equal than "MissingMsg_Default_Time" parameter, then IPC

- set DTC as "validated"
- IPC shall Consider STATUS_C_CAN.EngineSts=On

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_C_CAN message is correctly received, then IPC shall:

- set DTC as "devalidated"
- IPC
- IPC shall Consider STATUS_C_CAN.EngineSts received on CAN

1.13.2.3.2 STATUS_B_TCM_MTA_DCTM missing message description

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_B_TCM_MTA_DCTM is not received for a time greater or equal than "MissingMsg_Default_Time" parameter, then IPC

- set DTC as "validated"
- IPC shall send GearInsertNAndPressBrakeToStart.Req = Not_Active and EcoPower.Req= No_programs_available_selected

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If STATUS_B_TCM_MTA_DCTM message is correctly received, then IPC shall:

- set DTC as "devalidated"
- IPC shall send GearInsertNAndPressBrakeToStart.Req and EcoPower.Req coherent with the strategies described in this document.
- IPC shall Consider STATUS_C_CAN.EngineSts received on CAN

1.13.2.3.3 EDR_INFO missing message description

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If EDR_INFO is not received for a time greater or equal than "MissingMsg_Default_Time" parameter, then IPC

- set DTC as "validated"
- IPC shall consider EDR_INFO.BrakePedalSwitchNOSts=Not_active

In the following "Detection Condition":

- Keyon
- Keyon_engineON

If EDR_INFO message is correctly received, then IPC shall:

- set DTC as "devalidated"
- IPC shall Consider EDR_INFO.BrakePedalSwitchNOSTs received on CAN

1.14 Function properties

1.14.1 Configuration parameters

1.14.1.1 Configuration parameters table

Parameter	Description of the parameter in detail	Node	Implementation*
Rthreshold	Vehicle speed threshold to insert the Reverse	MTA	SUPPLIER EOL
Forwardthreshold	Vehicle speed threshold to insert the Forward gear	MTA	SUPPLIER EOL
T_SHOW_OFF_MTA	Time at keyon-->keyoff where the MTA shall guarantee to send on CAN net updated info	MTA	SUPPLIER EOL
Tmessage	Time to drive the message to IPC	MTA	SUPPLIER EOL
T_SHOW_OFF_IPC	Time at keyon-->keyoff where the IPC shall guarantee to show the requested indication	IPC	SUPPLIER EOL
MissingMsg_Default_Time	Time to recognize the missing message	IPC	SUPPLIER EOL
MissingMsg_Default_Time	Time to recognize the missing message	MTA	SUPPLIER EOL
MissingMsg_Default_Time	Time to recognize the missing message	BCM	SUPPLIER EOL
TPBM_threshold	Vehicle Speed Threshold to require to insert TPBM	MTA	SUPPLIER EOL
Tdebounce	timing of debounce and filtering	MTA	SUPPLIER EOL
Tstuck	Timing to recognize the drive mode button plausability fault	MTA	SUPPLIER EOL
MTA_CRC_MC_wrong_timer	Timer to recognize CRC/MC wrong	MTA	SUPPLIER EOL
T_Gear_Mismatch	timer to wait to recognize mismatch	MTA	SUPPLIER EOL

*This decision will be taken according with the supplier.

Possible values

- TBD
- PROXI
- FIAT EOL
- SUPPLIER EOL
- PART NUMBER

1.14.1.2 Configuration parameters description

Rthreshold: calibrable value: first trial value = 3kmh

Forwardthreshold: calibrable value: first trial value = -3kmh

T_SHOW_OFF_MTA: calibrable value: first trial value = 8 sec

Tmessage: calibrable value: first trial value = 10 sec

T_SHOW_OFF_IPC: calibrable value: first trial value = 7 sec

T_SHOW_OFF_BCM: calibrable value: first trial value = 8 sec

MissingMsg_Default_Time: calibrable value: first trial value = 2,5 sec

TPBM_threshold: calibrable value: first trial value = 4km/h

Tdebounce = calibrable value: first trial value = 60 msec

Tstuck = calibrable value: first trial value = 20 sec

MTA_CRC_MC_wrong_timer = calibrable value: first trial value = 3

T_Gear_Mismatch = calibrable value: first trial value = 500ms

1.14.2 Network wake up

1.14.2.1 Network wakeup table

Input (Component name)	Event description	Node	Ext	Int
Not Applicable				

1.15 Test pattern

1.16 Notes

1.16.1 Acronyms

1.16.2 Reference documents

All the Indication Strategy are described in the shift lever and MTA finalized functionality specification.